Louis Bensard

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SUMMARY

Machine learning engineer and co-founder & CTO with 7+ years taking ML from research to production. Designed and trained a custom transformer forecasting system (PatchTST, hurdle heads, multi-seed ensembles) in PyTorch, built the GPU training and inference infrastructure on AWS and the data platform behind it, and shipped it to enterprise customers as the company's first engineer. Grounded in mathematics and statistics, with 5 years of Bayesian and predictive modeling for Fortune 500 clients and experience leading data science teams of 9+.

EXPERIENCE

SIGHTLINE New York, NY
Co-founder & CTO (previously Founding Head of AI & ML) *Aug 2025 – Present*
First engineer. Built the company's AI system, data platform, product and engineering team from scratch, now in production for 10+ enterprise customers.

- **Forecasting model:** Designed and implemented a PatchTST transformer from scratch in PyTorch (patch tokenization, learned positional and per-series embeddings, multi-head encoder) with a hurdle architecture for zero-inflated demand. Replaced SARIMAX baselines; sub-5% wMAPE across thousands of series.
- **Training and evaluation:** Built the training stack: AdamW with LR scheduling and early stopping, 5-seed training with top-k ensembling, point-in-time backtesting, and an evaluation framework (MAPE, wMAPE, MAE) that gates every model change.
- **Hierarchical forecasting:** Built a second-stage allocation model that disaggregates store-level forecasts to item level, plus the data-preparation pipeline behind training: imputation, cold-start handling and store-migration logic.
- **GPU training platform:** Built model training on AWS Batch: CUDA images in ECR, scheduled retraining via Lambda, versioned model artifacts in S3, inference-only daily runs and experiment tracking in Weights & Biases.
- **Decision intelligence (SCAI):** Invented and built Supply Continuity AI, a recommendation engine new to the category that turns forecasts and inventory projections into scored escalations with a next-best action, recommended order quantities and dates.
- **Data platform:** Designed the event-driven ingestion platform (AWS Transfer Family, S3, SQS, Lambda, ECS, EC2): a config-driven mapper normalizing 7 distributor formats into one canonical schema, idempotent loads into PostgreSQL, and Grafana monitoring with alerting. Millions of rows processed daily.
- **Client implementations:** Technical lead for 10+ enterprise implementations end to end: scoped data integrations with each client's distributors and IT teams, built the connections and mappings, and managed the lifecycle through go-live.
- **Product and team:** Architected the multi-tenant product (Next.js, TypeScript, Prisma, Vercel) with org isolation and passwordless RDS access via IAM; built the engineering team and standards: hiring, code review, testing, CI/CD and documentation.

  Tech: Python, PyTorch, scikit-learn, pandas, SQLAlchemy, PostgreSQL, AWS (Batch, S3, SQS, Lambda, ECS, Transfer Family, RDS, Secrets Manager), Docker, Weights & Biases, Grafana, GitHub Actions, TypeScript, Next.js, Prisma

EKIMETRICS New York, NY
Tech Lead, Machine Learning *Jul 2024 – Aug 2025*

- **Bayesian modeling:** Built Bayesian marketing-mix models in PyMC with MCMC inference, estimating channel-level effects and ROI from time series KPI data for Fortune 500 retail and beauty clients.
- **Forecasting and deployment:** Built time series forecasting models (PyMC, scikit-learn, XGBoost) and reusable Python/SQL modeling frameworks, taking them from scoping through deployment; $20M+ in measured client impact.
- **Leadership:** Led a team of 9+ data scientists; primary technical contact for 10+ strategic accounts, presenting results to C-level executives.

STRATOS AVIATION New York, NY
Founder & Lead AI/ML Engineer *Aug 2023 – Dec 2024*
A "Strava for pilots" iOS app, live on the App Store. Designed, built and shipped solo, from concept to launch in 8 months.

- **Recommender system:** Built personalized airport and route recommendations from nearest-neighbor retrieval over flight time series and geospatial features across 100K airports and 10M+ points of interest, with on-device inference via Core ML.

- **LLM fine-tuning:** Curated the training data and fine-tuned an OpenAI model into a pilot assistant that gives flight-performance recommendations.
- **ML pipelines and security:** Automated preprocessing, feature selection (Boruta, RFE), evaluation and Core ML export; built end-to-end encrypted messaging between pilots (RSA-2048, AES-256).

IPSOS MMA New York, NY
Tech Lead, Machine Learning *Jul 2019 – Jun 2023*
Promoted three times, Data Scientist (ML Engineering) to Tech Lead

- **Bayesian modeling:** Built Bayesian and hierarchical marketing-mix models (PyMC3, MCMC inference) on time series KPI data for Fortune 500 clients.
- **Predictive modeling at scale:** Developed and deployed hundreds of models, from gradient boosting to neural networks (XGBoost, TensorFlow, scikit-learn), on 10B+ data points; automated the team's feature selection, Bayesian hyperparameter optimization and evaluation pipelines.
- **Interpretability:** Explained model behavior to C-suite executives with SHAP, LIME and interpretable models, on projects worth $4–6M a year.

PANASONIC AVIONICS CORPORATION Fullerton, CA
Machine Learning Engineer (internship) *Jan 2019 – May 2019*

- Built an in-flight entertainment recommendation engine (scikit-learn, TensorFlow) predicting passenger viewing preferences with 85%+ accuracy on a 10M-row dataset; led a 7-person team.

EDUCATION

California State University, Fullerton 2019
M.S. Statistics, summa cum laude (4.0 GPA, top of class, Phi Kappa Phi). Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Bayesian Analysis, Probability Theory.

Sorbonne Université, Paris 2017
B.S. Mathematics and B.S. Computer Science (two degrees), summa cum laude.

SKILLS

Deep learning and ML: Transformers (PatchTST, attention, positional and series embeddings), hurdle and zero-inflated models, multi-seed ensembling, time series forecasting, hierarchical forecasting, recommender systems, Bayesian inference (MCMC), gradient boosting, LLM fine-tuning, model evaluation and backtesting, interpretability (SHAP)

ML infrastructure: PyTorch, GPU training (CUDA, AWS Batch), experiment tracking (Weights & Biases), model versioning, training and inference pipelines, Docker, CI/CD (GitHub Actions)

Data and cloud: Python, SQL, PostgreSQL, pandas, SQLAlchemy, Alembic, AWS (S3, SQS, Lambda, ECS, Batch, Transfer Family, RDS, IAM, Secrets Manager), Grafana

Product: TypeScript, Next.js, React, Prisma, Vercel, Playwright, Swift, Core ML

Mathematics: Probability theory, linear algebra, Bayesian statistics, optimization, statistical modeling

PASSIONS

Licensed private pilot · Competitive rock climber (France, national level) · Overland travel across Africa · English, French (native), Spanish